We create chemistry

Safety Data Sheet
n-BUTYL ACETATE

1. Identification

Product identifier used on the label

n-BUTYL ACETATE

Recommended use of the chemical and restriction on use

Recommended use*: solvent(s)
Recommended use*: industrial chemicals; solvent(s)
Unsuitable for use: Not intended for sale to or use by the general public.

* The “Recommended use” identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:
BASF CORPORATION
100 Park Avenue
Florham Park, NJ 07932, USA

Telephone: +1 973 245-6000

Emergency telephone number
24 Hour Emergency Response Information
CHEMTREC: 1-800-424-9300
BASF HOTLINE: 1-800-832-HELP (4357)

Other means of identification
Chemical family: acetate

2. Hazards Identification

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

Flam. Liq.	3	Flammable liquids
STOT SE	3 (May cause drowsiness and dizziness.)	Specific target organ toxicity — single exposure

Aquatic Acute	3	Hazardous to the aquatic environment - acute

Label elements

Pictogram:

Signal Word:
Warning

Hazard Statement:

H226	Flammable liquid and vapour.
H336	May cause drowsiness or dizziness.
H402	Harmful to aquatic life.

Precautionary Statements (Prevention):

P271	Use only outdoors or in a well-ventilated area.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P280	Wear protective gloves and eye protection or face protection.
P261	Avoid breathing mist or vapour or spray.
P243	Take action to prevent static discharges.
P273	Avoid release to the environment.
P241	Use explosion-proof electrical, ventilating and lighting equipment.
P240	Ground and bond container and receiving equipment.
P242	Use non-sparking tools.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P303 + P361 + P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower.
P370 + P378	In case of fire: Use alcohol-resistant foam, carbon dioxide, dry powder or water spray for extinction.

Precautionary Statements (Storage):

P233	Keep container tightly closed.
P403 + P235	Store in a well-ventilated place. Keep cool.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents/container in accordance with local regulations.

Hazards not otherwise classified

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
See section 12 - Results of PBT and vPvB assessment.

Labeling of special preparations (GHS):
Repeated exposure may cause skin dryness or cracking.

3. Composition / Information on Ingredients

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

n-Butyl acetate
CAS Number: 123-86-4
Content (W/W): >= 99.5 - <= 99.8%
Synonym: Essigsäure-n-butylester

n-butanol
CAS Number: 71-36-3
Content (W/W): >= 0.15 - <= 0.15%
Synonym: 1-Butanol; n-Butanol

4. First-Aid Measures

Description of first aid measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

If on skin:
Wash thoroughly with soap and water

If in eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

If swallowed:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Overexposure may cause:, unconsciousness, vomiting, weakness, coordination disorder, nausea, diarrhea, coughing, headache
Hazards: Danger of drowsiness and dizziness.
Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
(Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Note to physician
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
dry powder, water spray, carbon dioxide, alcohol-resistant foam

Unsuitable extinguishing media for safety reasons:
water jet

Additional information:
Use extinguishing measures to suit surroundings.

Special hazards arising from the substance or mixture

Hazards during fire-fighting:
Flammable liquid Cool endangered containers with water-spray. See SDS section 7 - Handling and storage.

Advice for fire-fighters

Protective equipment for fire-fighting:
Firefighters should be equipped with self-contained breathing apparatus and turn-out gear. Special protective equipment for firefighters

Further information:
Evacuate area of all unnecessary personnel. Fight fire from maximum distance.

Extend fire extinguishing measures to the surroundings. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

Impact Sensitivity:
Remarks: Based on the chemical structure there is no shock-sensitivity.

6. Accidental release measures

Further accidental release measures:
High risk of slipping due to leakage/spillage of product.

Release of substance/product can cause fire or explosion. Shut off or stop source of leak. Shut off or stop released substance/product under safe conditions.

Pack in tightly closed containers for disposal.

Personal precautions, protective equipment and emergency procedures

Handle in accordance with good industrial hygiene and safety practice.

Avoid all sources of ignition: heat, sparks, open flame. Use antistatic tools.

Environmental precautions

Discharge into the environment must be avoided.

Methods and material for containment and cleaning up

Pick up with suitable appliance and dispose of. Spills should be contained, solidified, and placed in suitable containers for disposal. Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practice.

Ground and/or bond all equipment to prevent electrostatic charges. Avoid all sources of ignition: heat, sparks, open flame.

Protection against fire and explosion:
Avoid all sources of ignition: heat, sparks, open flame. Ground all transfer equipment properly to prevent electrostatic discharge.

Conditions for safe storage, including any incompatibilities

No applicable information available.

Further information on storage conditions: Keep container tightly closed in a cool, well-ventilated place.

8. Exposure Controls/Personal Protection

Components with occupational exposure limits

n-butanol	ACGIH, US:	TWA value 20 ppm ;
	OSHA Z1:	PEL 100 ppm 300 mg/m3 ;
n-Butyl acetate	ACGIH, US:	STEL value 150 ppm ;
	ACGIH, US:	TWA value 50 ppm ;
	OSHA Z1:	PEL 150 ppm 710 mg/m3 ;

Advice on system design:
No applicable information available.

Personal protective equipment

Respiratory protection:
Wear a NIOSH-certified (or equivalent) organic vapour respirator.

Hand protection:
Chemical resistant protective gloves, Protective glove selection must be based on the user's assessment of the workplace hazards.

Eye protection:
Safety glasses with side-shields.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boots, chemical-protection suit.

General safety and hygiene measures:
Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment. Avoid inhalation.

9. Physical and Chemical Properties

Form:	liquid	
Odour:	fruity	
Odour threshold:	not determined	
Colour:	colourless	
pH value:	not applicable	
Melting point:	-78 °C Literature data.	
Freezing point:	No data available.	
Boiling point:	124 - 126.5 °C (1,013 hPa) Literature data.	
Boiling range:	No data available.	
Sublimation point:	No applicable information available.	
Flash point:	27 °C	(Directive 92/69/EEC, A.9, closed cup)
Flammability:	Flammable.	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling. The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Autoignition:	415 °C	(DIN 51794)
Vapour pressure:	15 hPa (20 °C) Extrapolated value, static	(measured)
Density:	0.8812 g/cm3 (20 °C) 0.54 g/cm3 (55 °C)	(DIN 51757)
Relative density:	0.8813 (20 °C) Literature data.	
Vapour density:	4 (20 °C) Heavier than air.	(calculated)
Partitioning coefficient n-octanol/water (log Pow):	2.3 (25 °C)	(OECD Guideline 117)
Self-ignition temperature:	Based on its structural properties the product is not classified as self-igniting.	
Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.	
Viscosity, kinematic:	0.83 mm2/s (20 °C)	(OECD Guideline 114)
Particle size:	The substance / product is marketed or used in a non solid or granular form.	

Solubility in water:	5.3 g/l (20 °C) pH 6
Solubility (quantitative):	No applicable information available.
Solubility (qualitative):	soluble solvent(s): organic solvents,
Molar mass:	116.16 g/mol
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.

10. Stability and Reactivity

Reactivity
When heated can give off ignitable vapours.

Corrosion to metals:
No corrosive effect on metal.

Oxidizing properties:
not fire-propagating

Formation of flammable gases:	Remarks:	Forms no flammable gases in the presence of water.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
Reacts with strong oxidizing agents.

Conditions to avoid
Avoid all sources of ignition: heat, sparks, open flame.

Incompatible materials
strong oxidizing agents

Hazardous decomposition products

Decomposition products:
Hazardous decomposition products: No hazardous decomposition products if stored and handled as prescribed/indicated.

Thermal decomposition:
No decomposition if stored and handled as prescribed/indicated.

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity
Assessment of acute toxicity: Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact.

Oral
Type of value: LD50
Species: rat (male/female)
Value: 10,736 mg/kg (other)

Inhalation
Type of value: LC50
Species: rat (male/female)
Value: > 21.1 mg/l (OECD Guideline 403)
Exposure time: 4 h
The vapour was tested.

Type of value: LC0
Species: rat (male/female)
Value: > 38.32 mg/l > 8000 ppm (other)
Exposure time: 6 h
The vapour was tested.

Dermal
Type of value: LD50
Species: rabbit (male/female)
Value: > 14,000 mg/kg (other)

Assessment other acute effects
Assessment of STOT single:
Possible narcotic effects (drowsiness or dizziness).

Irritation / corrosion
Assessment of irritating effects: Not irritating to the skin. Not irritating to the eyes.
Not irritating to the skin. May cause slight irritation to the eyes.

Skin
Species: rabbit
Result: non-irritant
Method: OECD Guideline 404

Eye
Species: rabbit
Result: non-irritant
Method: OECD Guideline 405

Sensitization
Assessment of sensitization: Skin sensitizing effects were not observed in animal studies.

Guinea pig maximization test
Species: guinea pig
Result: Non-sensitizing.
Method: other

Aspiration Hazard
No aspiration hazard expected.

Chronic Toxicity/Effects

Repeated dose toxicity
Assessment of repeated dose toxicity: After repeated exposure the prominent effect is local irritation.

Genetic toxicity
Assessment of mutagenicity: No mutagenic effect was found in various tests with microorganisms and mammalian cell culture. The substance was not mutagenic in studies with mammals.
The substance was not mutagenic in bacteria. The substance was not genotoxic in mammalian cell culture. The substance was not genotoxic in a test with mammals. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.
Genetic toxicity in vitro: other Ames-test S. typhimurium, E. coli:with and without metabolic activation negative
other Cytogenetic assay CHL cells:without metabolic activation negative
OECD Guideline 476 HGPRT assay V79 cells:with and without metabolic activation negative
Genetic toxicity in vivo: OECD Guideline 474 Micronucleus assay mouse (NMRI) (male/female) negative
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Carcinogenicity
Assessment of carcinogenicity: Study does not need to be conducted.

Reproductive toxicity
Assessment of reproduction toxicity: The results of animal studies gave no indication of a fertility impairing effect.

Teratogenicity
Assessment of teratogenicity: Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

Experiences in humans
High concentrations have a narcotizing effect. Prolonged contact can result in drying of the skin.

Other Information
Has a degreasing effect on skin.

12. Ecological Information

Toxicity

Aquatic toxicity
Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish
LC50 (96 h) 18 mg/l, Pimephales promelas (Fish test acute, Flow through.)
The statement of the toxic effect relates to the analytically determined concentration.

Aquatic invertebrates

EC50 (48 h) 44 mg/l, Daphnia sp. (Daphnia test acute, static)
Nominal concentration.

Aquatic plants
EC50 (72 h) 397 mg/l (growth rate), Pseudokirchneriella subcapitata (DIN 38412 Part 9)
The statement of the toxic effect relates to the analytically determined concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Chronic toxicity to fish
No data available.

Chronic toxicity to aquatic invertebrates
No observed effect concentration (21 d) 23 mg/l, Daphnia magna (OECD Guideline 211, semistatic)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Assessment of terrestrial toxicity
No toxic effects have been observed in studies with terrestric plants.

Soil living organisms

Toxicity to soil dwelling organisms:
No data available.

Toxicity to terrestrial plants

EC50 (14 d) > 1,000 mg/kg > 1.000 mg/kg, Lactuca sativa (OECD Guideline 208)

Other terrestrial non-mammals
No data available.

Microorganisms/Effect on activated sludge

Toxicity to microorganisms
internal method aquatic
Protozoa/EC50 (40 h): 356 mg/l

Persistence and degradability

Assessment biodegradation and elimination (H2O)
Readily biodegradable (according to OECD criteria).

Elimination information

80 % BOD of the ThOD (5 d) (OECD 301D; 92/69/EWG, C.4-E) (aerobic, municipal sewage treatment plant effluent)

Assessment of stability in water
In contact with water the substance will hydrolyse slowly.

Information on Stability in Water (Hydrolysis)
$t_{1/2}$ 782 d, (calculated, pH 7)

Bioaccumulative potential

Assessment bioaccumulation potential
No significant accumulation in organisms is expected as a result of the distribution coefficient of n-octanol/water (log Pow).

Bioaccumulation potential
No data available.

Mobility in soil

Assessment transport between environmental compartments
The substance will slowly evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is not expected.

Additional information

Adsorbable organically-bound halogen(AOX):
This product contains no organically-bound halogen.

13. Disposal considerations

Waste disposal of substance:
Dispose of in accordance with national, state and local regulations.

Container disposal:
Empty containers with less than 1 inch of residue may be landfilled at a licensed facility.

Flammable vapors may exist in containers in which residues of this product remain. DO NOT PRESSURIZE, CUT, WELD, BRAZE, SOLDER, DRILL, GRIND, OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION: THEY MAY EXPLODE AND CAUSE INJURY OR DEATH.

RCRA: D001

14. Transport Information

Land transport
USDOT

Hazard class:	3
Packing group:	III
ID number:	UN 1123
Hazard label:	3
Proper shipping name:	BUTYL ACETATES

Sea transport
IMDG

Hazard class:	3
Packing group:	III
ID number:	UN 1123
Hazard label:	3
Marine pollutant:	NO
Proper shipping name:	BUTYL ACETATES

Air transport
IATA/ICAO

Hazard class:	3

Packing group:	III
ID number:	UN 1123
Hazard label:	3
Proper shipping name:	BUTYL ACETATES

15. Regulatory Information

Federal Regulations

Registration status:
Chemical TSCA, US released / listed

EPCRA 311/312 (Hazard categories): Refer to SDS section 2 for GHS hazard classes applicable for this product.

CERCLA RQ	CAS Number	Chemical name
5000 LBS	123-86-4	n-Butyl acetate

State regulations

State RTK	CAS Number	Chemical name
NJ	123-86-4	n-Butyl acetate
PA	123-86-4	n-Butyl acetate

NFPA Hazard codes:
Health: 1 Fire: 3 Reactivity: 0 Special:

HMIS III rating
Health: 1 Flammability: 3 Physical hazard: 0

Assessment of the hazard classes according to UN GHS criteria (most recent version):

Flam. Liq.	3	Flammable liquids
STOT SE	3 (May cause drowsiness and dizziness.)	Specific target organ toxicity — single exposure
Aquatic Acute	3	Hazardous to the aquatic environment - acute

16. Other Information

SDS Prepared by:
BASF NA Product Regulations
SDS Prepared on: 2023/12/19

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